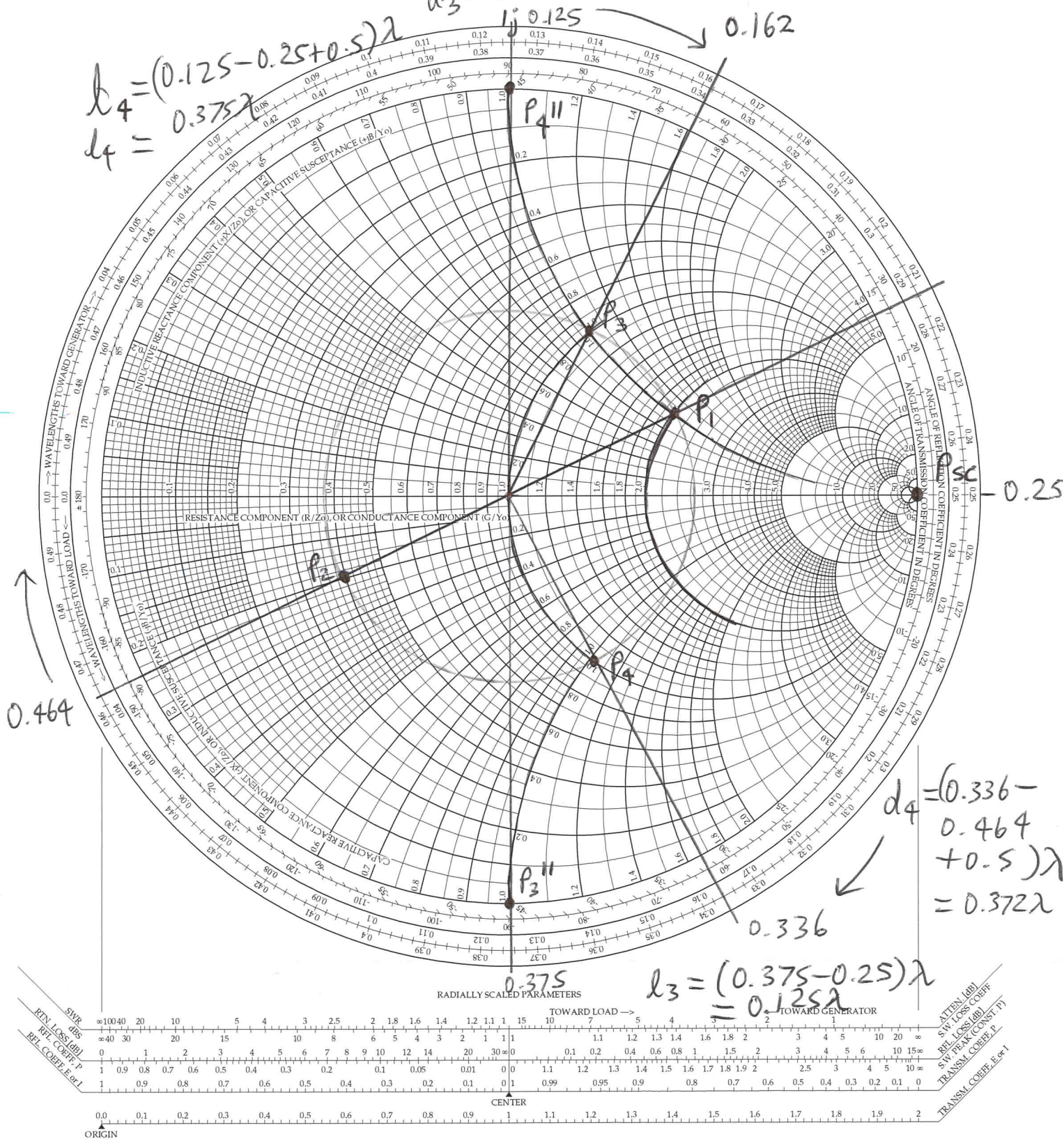


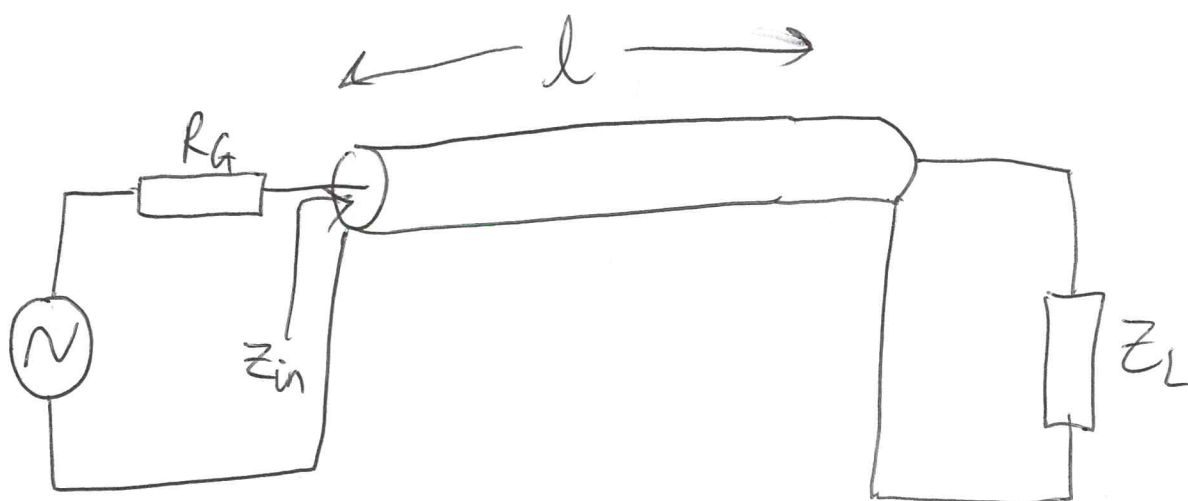
$$Z_L = 2 + j1$$

$$y_L = 0.4 - j0.21$$

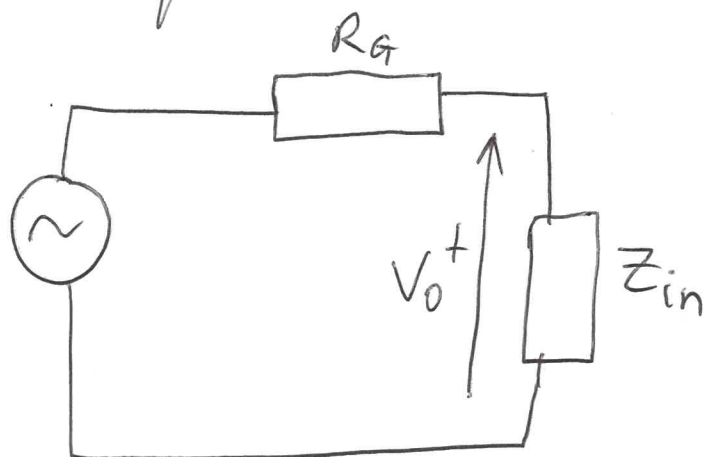
The Complete Smith Chart

$$d_3 = (0.162 - 0.464 + 0.5) \lambda = 0.198 \lambda$$



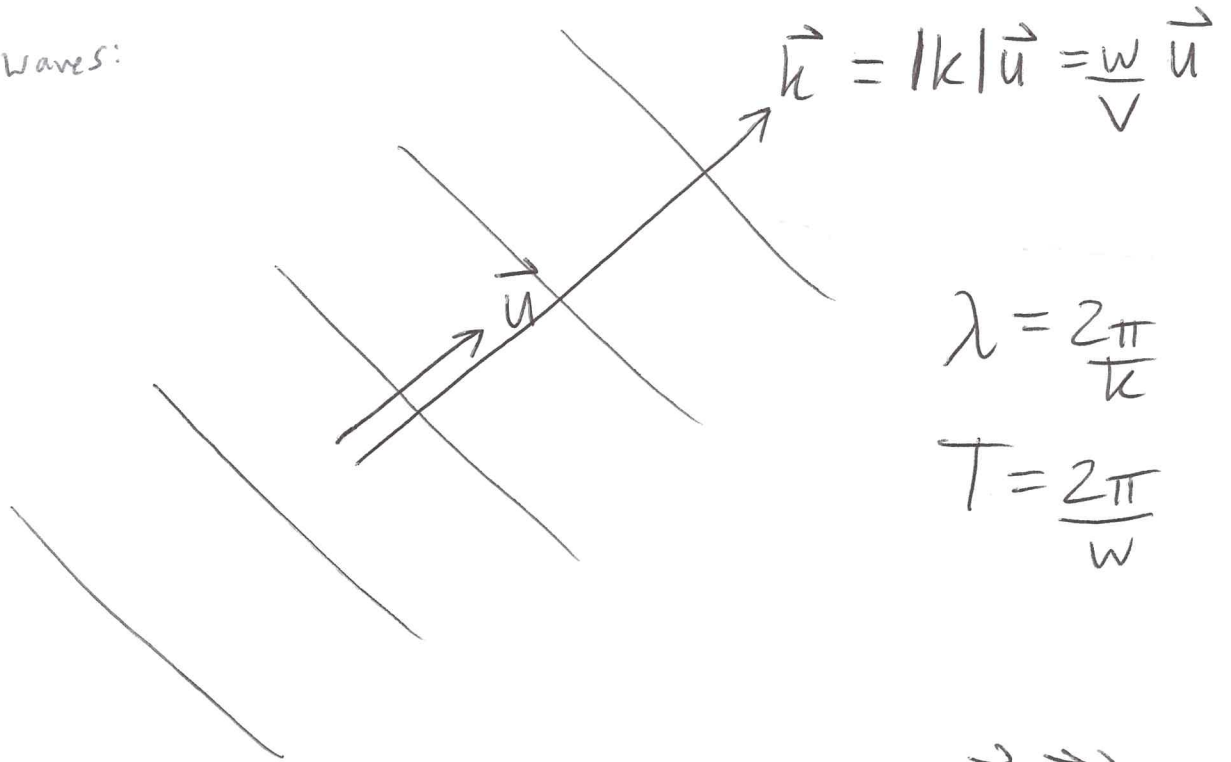


Equivalent circuit:



$$V(z = -l) = V_0^+ = V_G \frac{Z_{in}}{Z_{in} + R_G}$$

Plane waves:



$$\lambda = \frac{2\pi}{k}$$

$$T = \frac{2\pi}{\omega}$$

e.g. $E(\vec{r}, t) = E_0 \cos(\omega t \pm \vec{k} \cdot \vec{r})$

t : $\lambda = \frac{2\pi}{k}$

$\vec{r}$: $T = \frac{2\pi}{\omega}$

Coaxial cables

$$k = \beta = \frac{2\pi}{\lambda}$$

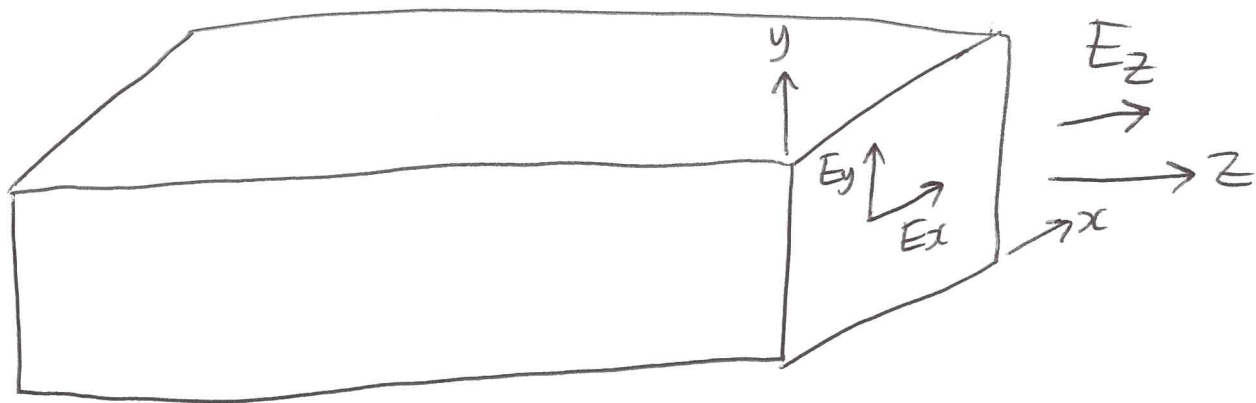
Wave guides

$$k = \frac{2\pi}{\lambda}$$

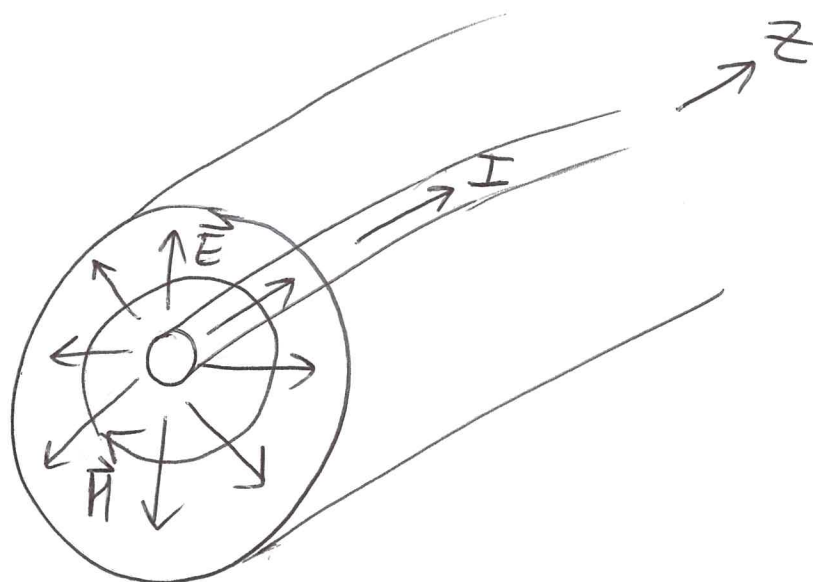
property of
medium

$$\beta \neq k$$

property of
medium +
structure

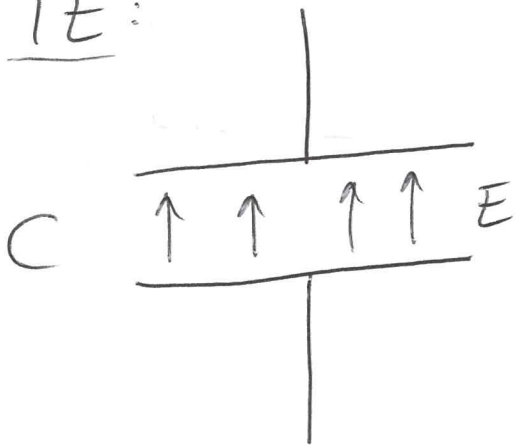


$$\vec{E}_{\perp} = \vec{E}_x + \vec{E}_y$$



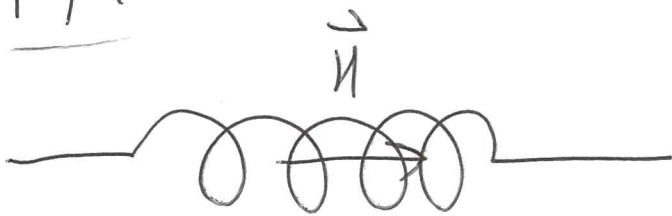
$$Z_{TE} > Z_0 = \sqrt{\frac{\mu}{\epsilon}}$$

TE:



$$Z = \frac{1}{j\omega C}$$

TM:



$$Z = j\omega L$$

$$Z_{TM} < Z_0$$

